

STA314 Tutorial 1

R Review and Linear Regression

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Quick R Review

Load and inspect the data

We will use the `Credit` dataset from the `ISLR2` package.

```
library(ISLR2)
data("Credit")
```

```
# First six observations
head(Credit)
```

```
##      Income Limit Rating Cards Age Education Own Student Married Region Balance
## 1  14.891  3606   283    2  34         11 No      No      Yes  South    333
## 2 106.025  6645   483    3  82         15 Yes    Yes    Yes   West    903
## 3 104.593  7075   514    4  71         11 No     No     No    West    580
## 4 148.924  9504   681    3  36         11 Yes    No     No    West    964
## 5  55.882  4897   357    2  68         16 No     No     Yes  South    331
## 6  80.180  8047   569    4  77         10 No     No     No    South   1151
```

```
# Variable names
names(Credit)
```

```
## [1] "Income" "Limit" "Rating" "Cards" "Age" "Education"
## [7] "Own" "Student" "Married" "Region" "Balance"
```

```
# Structure and variable types
str(Credit)
```

```
## 'data.frame': 400 obs. of 11 variables:
## $ Income : num 14.9 106 104.6 148.9 55.9 ...
## $ Limit : num 3606 6645 7075 9504 4897 ...
## $ Rating : num 283 483 514 681 357 569 259 512 266 491 ...
## $ Cards : num 2 3 4 3 2 4 2 2 5 3 ...
## $ Age : num 34 82 71 36 68 77 37 87 66 41 ...
## $ Education: num 11 15 11 11 16 10 12 9 13 19 ...
## $ Own : Factor w/ 2 levels "No","Yes": 1 2 1 2 1 1 2 1 2 2 ...
## $ Student : Factor w/ 2 levels "No","Yes": 1 2 1 1 1 1 1 1 1 2 ...
## $ Married : Factor w/ 2 levels "No","Yes": 2 2 1 1 2 1 1 1 1 2 ...
## $ Region : Factor w/ 3 levels "East","South",...: 2 3 3 3 2 2 1 3 2 1 ...
## $ Balance : num 333 903 580 964 331 ...
```

```
# Number of observations and variables
dim(Credit)
```

```
## [1] 400 11
```

Quick check: What information does `str(Credit)` provide that `head(Credit)` does not?

Access variables and calculate summaries

The `$` operator extracts one variable from a data frame.

```
# Extract Income
```

```
Credit$Income
```

```
## [1] 14.891 106.025 104.593 148.924 55.882 80.180 20.996 71.408 15.125
## [10] 71.061 63.095 15.045 80.616 43.682 19.144 20.089 53.598 36.496
## [19] 49.570 42.079 17.700 37.348 20.103 64.027 10.742 14.090 42.471
## [28] 32.793 186.634 26.813 34.142 28.941 134.181 31.367 20.150 23.350
## [37] 62.413 30.007 11.795 13.647 34.950 113.659 44.158 36.929 31.861
## [46] 77.380 19.531 44.646 44.522 43.479 36.362 39.705 44.205 16.304
## [55] 15.333 32.916 57.100 76.273 10.354 51.872 35.510 21.238 30.682
## [64] 14.132 32.164 12.000 113.829 11.187 27.847 49.502 24.889 58.781
## [73] 22.939 23.989 16.103 33.017 30.622 20.936 110.968 15.354 27.369
## [82] 53.480 23.672 19.225 43.540 152.298 55.367 11.741 15.560 59.530
## [91] 20.191 48.498 30.733 16.479 38.009 14.084 14.312 26.067 36.295
## [100] 83.851 21.153 17.976 68.713 146.183 15.846 12.031 16.819 39.110
## [109] 107.986 13.561 34.537 28.575 46.007 69.251 16.482 40.442 35.177
## [118] 91.362 27.039 23.012 27.241 148.080 62.602 11.808 29.564 27.578
## [127] 26.427 57.202 123.299 18.145 23.793 10.726 23.283 21.455 34.664
## [136] 44.473 54.663 36.355 21.374 107.841 39.831 91.876 103.893 19.636
## [145] 17.392 19.529 17.055 23.857 15.184 13.444 63.931 35.864 41.419
## [154] 92.112 55.056 19.537 31.811 56.256 42.357 53.319 12.238 31.353
## [163] 63.809 13.676 76.782 25.383 35.691 29.403 27.470 27.330 34.772
## [172] 36.934 76.348 14.887 121.834 30.132 24.050 22.379 28.316 58.026
## [181] 10.635 46.102 58.929 80.861 158.889 30.420 36.472 23.365 83.869
## [190] 58.351 55.187 124.290 28.508 130.209 30.406 23.883 93.039 50.699
## [199] 27.349 10.403 23.949 73.914 21.038 68.206 57.337 10.793 23.450
## [208] 10.842 51.345 151.947 24.543 29.567 39.145 39.422 34.909 41.025
## [217] 15.476 12.456 10.627 38.954 44.847 98.515 33.437 27.512 121.709
## [226] 15.079 59.879 66.989 69.165 69.943 33.214 25.124 15.741 11.603
## [235] 69.656 10.503 42.529 60.579 26.532 27.952 29.705 15.602 20.918
## [244] 58.165 22.561 34.509 19.588 36.364 15.717 22.574 10.363 28.474
## [253] 72.945 85.425 36.508 58.063 25.936 15.629 41.400 33.657 67.937
## [262] 180.379 10.588 29.725 27.999 40.885 88.830 29.638 25.988 39.055
## [271] 15.866 44.978 30.413 16.751 30.550 163.329 23.106 41.532 128.040
## [280] 54.319 53.401 36.142 63.534 49.927 14.711 18.967 18.036 60.449
## [289] 16.711 10.852 26.370 24.088 51.532 140.672 42.915 27.272 65.896
## [298] 55.054 20.791 24.919 21.786 31.335 59.855 44.061 82.706 24.460
## [307] 45.120 75.406 14.956 75.257 33.694 23.375 27.825 92.386 115.520
## [316] 14.479 52.179 68.462 18.951 27.590 16.279 25.078 27.229 182.728
## [325] 31.029 17.765 125.480 49.166 41.192 94.193 20.405 12.581 62.328
## [334] 21.011 24.230 24.314 32.856 12.414 41.365 149.316 27.794 13.234
## [343] 14.595 10.735 48.218 30.012 21.551 160.231 13.433 48.577 30.002
## [352] 61.620 104.483 41.868 12.068 180.682 34.480 39.609 30.111 12.335
## [361] 53.566 53.217 26.162 64.173 128.669 113.772 61.069 23.793 89.000
## [370] 71.682 35.610 39.116 19.782 55.412 29.400 20.974 87.625 28.144
## [379] 19.349 53.308 115.123 101.788 24.824 14.292 20.088 26.400 19.253
## [388] 16.529 37.878 83.948 135.118 73.327 25.974 17.316 49.794 12.096
## [397] 13.364 57.872 37.728 18.701
```

```
# First five Income values
Credit$Income[1:5]
```

```
## [1] 14.891 106.025 104.593 148.924 55.882
```

```
# Summary statistics
mean(Credit$Income)
```

```
## [1] 45.21889
```

```
mean(Credit$Balance)
```

```
## [1] 520.015
```

```
max(Credit$Income)
```

```
## [1] 186.634
```

Before running the following commands, predict what each will return.

```
Credit$Balance[1:3]
```

```
## [1] 333 903 580
```

```
mean(Credit$Rating)
```

```
## [1] 354.94
```

Logical conditions and subsetting

```
# Logical condition
Credit$Student == "Yes"
```

```
## [1] FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE
## [13] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [25] FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE
## [37] FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE TRUE TRUE
## [49] FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [61] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [73] FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [85] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [97] TRUE FALSE TRUE FALSE FALSE TRUE TRUE FALSE FALSE FALSE FALSE FALSE
## [109] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [121] FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE TRUE FALSE TRUE FALSE
## [133] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE
## [145] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE
## [157] FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [169] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [181] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE
## [193] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE
## [205] FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [217] FALSE FALSE TRUE FALSE TRUE FALSE TRUE FALSE FALSE FALSE FALSE FALSE
## [229] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [241] FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE TRUE TRUE FALSE FALSE
## [253] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE
## [265] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE TRUE FALSE FALSE
## [277] FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [289] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [301] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE
```

```
## [313] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE
## [325] TRUE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE
## [337] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [349] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [361] FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE
## [373] TRUE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE
## [385] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [397] FALSE FALSE FALSE FALSE
```

```
# Select all student rows
```

```
Credit[Credit$Student == "Yes", ]
```

##	Income	Limit	Rating	Cards	Age	Education	Own	Student	Married	Region	Balance
## 2	106.025	6645	483	3	82	15	Yes	Yes	Yes	West	903
## 10	71.061	6819	491	3	41	19	Yes	Yes	Yes	East	1350
## 27	42.471	3625	289	6	44	12	Yes	Yes	No	South	654
## 36	23.350	2558	220	3	49	12	Yes	Yes	No	South	419
## 42	113.659	7659	538	2	66	15	No	Yes	Yes	East	1155
## 47	19.531	5043	376	2	64	16	Yes	Yes	Yes	West	1241
## 48	44.646	4431	320	2	49	15	No	Yes	Yes	South	797
## 50	43.479	4569	354	4	49	13	No	Yes	Yes	East	902
## 77	30.622	3293	251	1	68	16	No	Yes	No	South	532
## 97	14.312	5382	367	1	59	17	No	Yes	No	West	1380
## 99	36.295	2963	241	2	68	14	Yes	Yes	No	East	375
## 102	17.976	2433	190	3	70	16	Yes	Yes	No	South	431
## 103	68.713	7582	531	2	56	16	No	Yes	No	South	1587
## 127	26.427	5533	433	5	50	15	Yes	Yes	Yes	West	1404
## 129	123.299	8376	610	2	89	17	No	Yes	No	East	1259
## 131	23.793	3821	281	4	56	12	Yes	Yes	Yes	East	868
## 141	39.831	6045	459	3	32	12	Yes	Yes	Yes	East	1425
## 153	41.419	2120	184	4	24	11	Yes	Yes	No	South	156
## 158	56.256	5521	406	2	72	16	Yes	Yes	Yes	South	1020
## 192	124.290	9560	701	3	52	17	Yes	Yes	No	West	1687
## 204	68.206	6784	499	5	40	16	Yes	Yes	No	East	1411
## 208	10.842	4391	358	5	37	10	Yes	Yes	Yes	South	1216
## 219	10.627	1647	149	2	71	10	Yes	Yes	Yes	West	195
## 221	44.847	5765	437	3	53	13	Yes	Yes	No	West	1246
## 223	33.437	6207	451	4	44	9	No	Yes	No	South	1549
## 243	20.918	1233	128	3	47	18	Yes	Yes	Yes	West	16
## 249	15.717	905	93	1	38	16	No	Yes	Yes	South	0
## 250	22.574	1551	134	3	43	13	Yes	Yes	Yes	South	98
## 262	180.379	9310	665	3	67	8	Yes	Yes	Yes	West	1050
## 273	30.413	3690	299	2	25	15	Yes	Yes	No	West	728
## 274	16.751	4706	353	6	48	14	No	Yes	No	West	1255
## 280	54.319	3063	248	3	59	8	Yes	Yes	No	South	269
## 311	33.694	4891	369	1	52	16	No	Yes	No	East	1036
## 321	16.279	1160	126	3	78	13	No	Yes	Yes	East	5
## 325	31.029	2863	223	2	66	17	No	Yes	Yes	West	415
## 331	20.405	4543	329	2	72	17	No	Yes	No	West	1054
## 366	113.772	6442	489	4	69	15	No	Yes	Yes	South	790
## 373	19.782	3782	293	2	46	16	Yes	Yes	No	South	840
## 374	55.412	5354	383	2	37	16	Yes	Yes	Yes	South	1003
## 383	24.824	5495	409	1	33	9	No	Yes	No	South	1352

```
# Select particular rows and columns
Credit[
  Credit$Student == "Yes",
  c("Income", "Balance")
]
```

```
##      Income Balance
## 2    106.025     903
## 10   71.061    1350
## 27   42.471     654
## 36   23.350     419
## 42  113.659    1155
## 47   19.531    1241
## 48   44.646     797
## 50   43.479     902
## 77   30.622     532
## 97   14.312    1380
## 99   36.295     375
## 102  17.976     431
## 103  68.713    1587
## 127  26.427    1404
## 129 123.299    1259
## 131  23.793     868
## 141  39.831    1425
## 153  41.419     156
## 158  56.256    1020
## 192 124.290    1687
## 204  68.206    1411
## 208  10.842    1216
## 219  10.627     195
## 221  44.847    1246
## 223  33.437    1549
## 243  20.918      16
## 249  15.717       0
## 250  22.574      98
## 262 180.379    1050
## 273  30.413     728
## 274  16.751    1255
## 280  54.319     269
## 311  33.694    1036
## 321  16.279       5
## 325  31.029     415
## 331  20.405    1054
## 366 113.772     790
## 373  19.782     840
## 374  55.412    1003
## 383  24.824    1352
```

```
# Examine categorical variables
levels(Credit$Student)
```

```
## [1] "No"  "Yes"
```

```
levels(Credit$Region)
```

```
## [1] "East" "South" "West"
```

```
table(Credit$Region)
```

```
##
```

```
## East South West
```

```
## 99 199 102
```

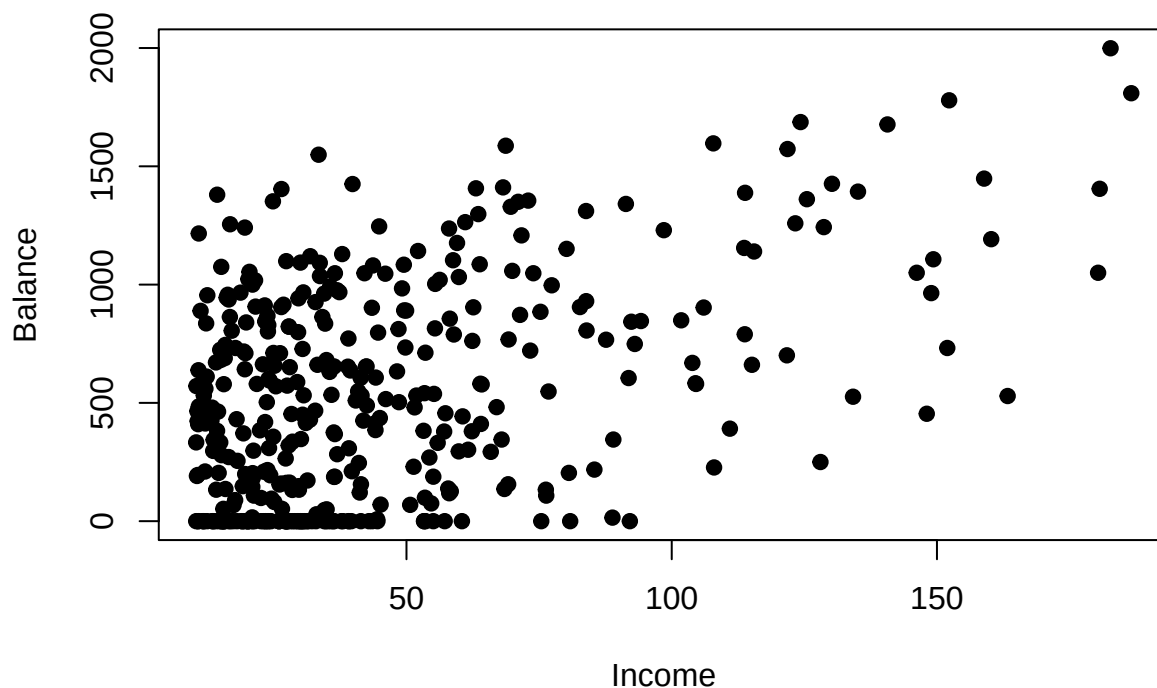
Recall that `data[rows, columns]` specifies the observations and variables to retain.

Guided Example: Credit Card Data

The `Credit` dataset contains simulated information on 400 individuals. We will use `Balance` as the response and consider predictors including `Income`, `Rating`, `Student`, and `Region`.

Explore Income and Balance

```
plot(  
  Credit$Income,  
  Credit$Balance,  
  xlab = "Income",  
  ylab = "Balance",  
  pch = 19  
)
```



Based only on this plot:

1. Does there appear to be a relationship between Income and Balance?
2. Does the relationship appear approximately linear?
3. Is there substantial variability around the relationship?

Fit a simple linear regression

Consider

$$\text{Balance}_i = \beta_0 + \beta_1 \text{Income}_i + \epsilon_i.$$

```
fit_slr <- lm(
  Balance ~ Income,
  data = Credit
)

summary(fit_slr)

##
## Call:
## lm(formula = Balance ~ Income, data = Credit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -803.64 -348.99  -54.42   331.75 1100.25
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  246.5148    33.1993   7.425  6.9e-13 ***
## Income        6.0484     0.5794  10.440 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 407.9 on 398 degrees of freedom
## Multiple R-squared:  0.215, Adjusted R-squared:  0.213
## F-statistic: 109 on 1 and 398 DF, p-value: < 2.2e-16
coef(fit_slr)

## (Intercept)      Income
##  246.514751    6.048363
```

The fitted model is approximately

$$\widehat{\text{Balance}} = 246.51 + 6.05(\text{Income}).$$

Because Income is recorded in thousands of dollars, the estimated slope means that an additional \$1,000 of Income is associated with an estimated 6.05 unit increase in Balance, on average.

Add the fitted line

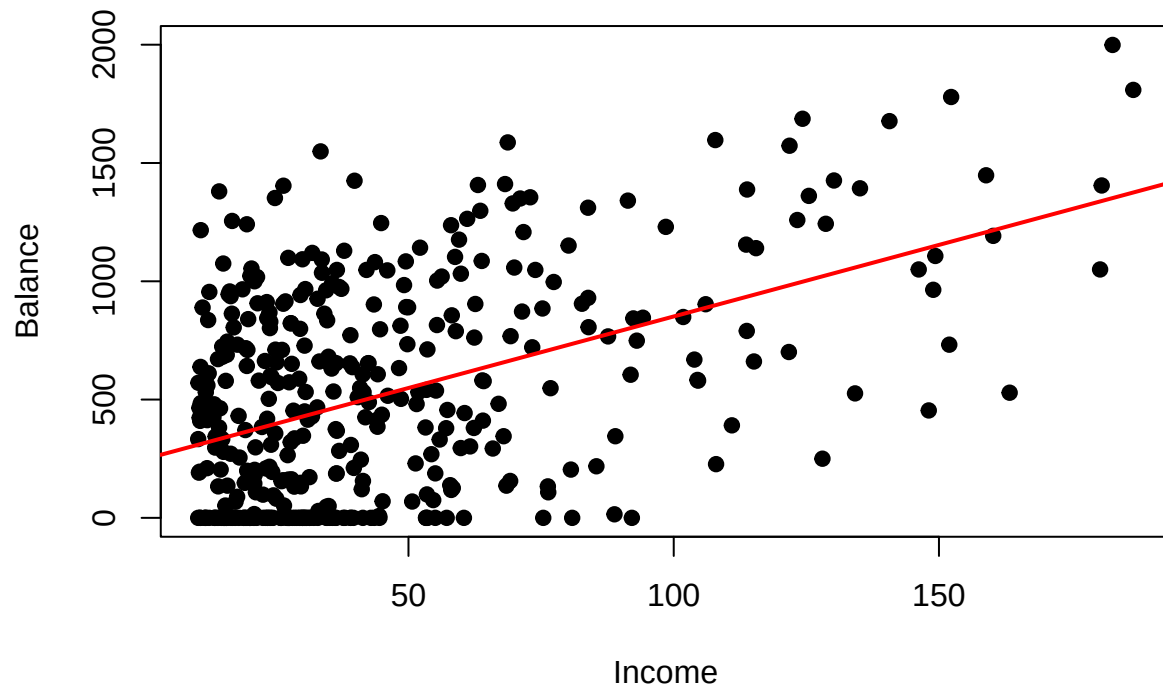
```
plot(
  Credit$Income,
  Credit$Balance,
  xlab = "Income",
  ylab = "Balance",
  pch = 19
)

abline(
```

```

fit_slr,
col = "red",
lwd = 2
)

```



Examine the residuals

```

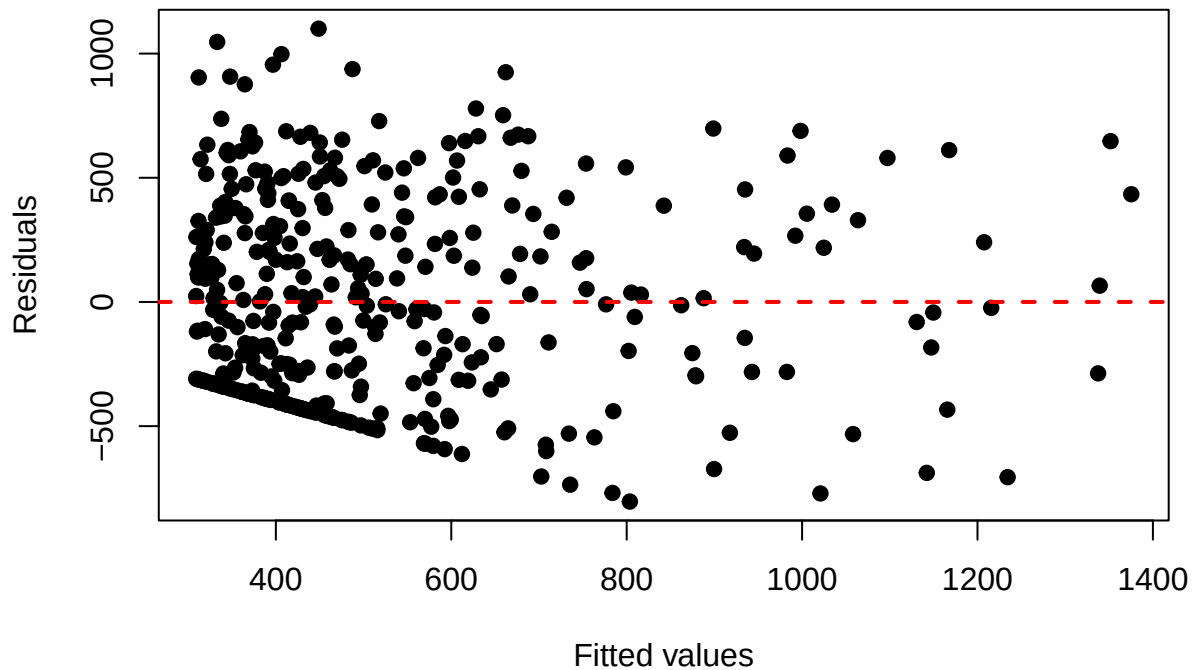
plot(
  fitted(fit_slr),
  residuals(fit_slr),
  xlab = "Fitted values",
  ylab = "Residuals",
  pch = 19
)

```

```

abline(
  h = 0,
  col = "red",
  lty = 2,
  lwd = 2
)

```



What pattern would we hope to observe if the linear model were appropriate? Do the residuals appear completely random?

Predictions and intervals

Suppose we want predictions for individuals with Income equal to 20, 40, and 80.

```
new_data <- data.frame(
  Income = c(20, 40, 80)
)
```

```
new_data
```

```
##   Income
## 1     20
## 2     40
## 3     80
```

Point predictions

```
predict(
  fit_slr,
  newdata = new_data
)
```

```
##           1           2           3
## 367.4820 488.4493 730.3838
```

Confidence intervals for the mean response

```
predict(
  fit_slr,
  newdata = new_data,
```

```

interval = "confidence",
level = 0.95
)

```

```

##          fit          lwr          upr
## 1 367.4820 318.1626 416.8014
## 2 488.4493 447.9191 528.9795
## 3 730.3838 674.0218 786.7459

```

Prediction intervals for new individuals

```

predict(
  fit_slr,
  newdata = new_data,
  interval = "prediction",
  level = 0.95
)

```

```

##          fit          lwr          upr
## 1 367.4820 -435.87183 1170.836
## 2 488.4493 -314.41290 1291.311
## 3 730.3838 -73.43312 1534.201

```

The prediction intervals are wider because they account for uncertainty about the mean response and individual variability around the regression line.

Multiple linear regression

Now model Balance using Income, Rating, and Student status.

```

fit_mlr <- lm(
  Balance ~ Income + Rating + Student,
  data = Credit
)

```

```
summary(fit_mlr)
```

```

##
## Call:
## lm(formula = Balance ~ Income + Rating + Student, data = Credit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -226.126  -80.445   -5.018    65.192   293.234
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -581.07889    13.83463  -42.00  <2e-16 ***
## Income       -7.87493     0.24021  -32.78  <2e-16 ***
## Rating        3.98747     0.05471   72.89  <2e-16 ***
## StudentYes   418.76028    17.23025   24.30  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 103.3 on 396 degrees of freedom

```

```
## Multiple R-squared:  0.9499, Adjusted R-squared:  0.9495
## F-statistic: 2502 on 3 and 396 DF,  p-value: < 2.2e-16
```

Answer the following:

1. Interpret the estimated Income coefficient while holding Rating and Student status fixed.
2. Interpret `StudentYes` while holding Income and Rating fixed.
3. Is `StudentYes` statistically significant at the 5% level?

Region as a categorical predictor

The `Region` variable has three levels.

```
levels(Credit$Region)
```

```
## [1] "East" "South" "West"
```

```
table(Credit$Region)
```

```
##
##   East South  West
##    99   199   102
```

Explicitly set East as the reference category and fit a model using `Region`.

```
Credit$Region <- relevel(
  Credit$Region,
  ref = "East"
)
```

```
fit_region <- lm(
  Balance ~ Region,
  data = Credit
)
```

```
summary(fit_region)
```

```
##
## Call:
## lm(formula = Balance ~ Region, data = Credit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -531.00 -457.08  -63.25   339.25  1480.50
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    531.00     46.32   11.464  <2e-16 ***
## RegionSouth   -12.50     56.68   -0.221    0.826
## RegionWest    -18.69     65.02   -0.287    0.774
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 460.9 on 397 degrees of freedom
## Multiple R-squared:  0.0002188, Adjusted R-squared:  -0.004818
## F-statistic: 0.04344 on 2 and 397 DF,  p-value: 0.9575
```

Interpretation:

- The intercept is the estimated mean Balance among individuals in the East.
- **RegionSouth** is the estimated difference in mean Balance between the South and East.
- **RegionWest** is the estimated difference in mean Balance between the West and East.

The p-values test whether each regional difference is zero. They do not test whether all three regional means are equal simultaneously.

Examine the model matrix

```
head(
  model.matrix(fit_region)
)
```

```
##      (Intercept) RegionSouth RegionWest
## 1             1             1           0
## 2             1             0           1
## 3             1             0           1
## 4             1             0           1
## 5             1             1           0
## 6             1             1           0
```

```
contrasts(Credit$Region)
```

```
##           South West
## East           0    0
## South          1    0
## West           0    1
```

Although Region has three categories, R creates only two dummy variables because East is the reference category.

Interaction between Income and Student status

The additive model assumes the same Income slope for students and non-students. To allow different slopes, fit an interaction model.

```
fit_interaction <- lm(
  Balance ~ Income * Student,
  data = Credit
)
```

```
summary(fit_interaction)
```

```
##
## Call:
## lm(formula = Balance ~ Income * Student, data = Credit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -773.39 -325.70  -41.13   321.65   814.04
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    200.6232     33.6984   5.953 5.79e-09 ***
## Income           6.2182      0.5921  10.502 < 2e-16 ***
## StudentYes     476.6758    104.3512   4.568 6.59e-06 ***
## Income:StudentYes -1.9992      1.7313  -1.155  0.249
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 391.6 on 396 degrees of freedom
## Multiple R-squared:  0.2799, Adjusted R-squared:  0.2744
## F-statistic: 51.3 on 3 and 396 DF,  p-value: < 2.2e-16
```

In an R formula,

```
Income * Student
```

is shorthand for

```
Income + Student + Income:Student
```

The Income slope for non-students is:

```
coef(fit_interaction)["Income"]
```

```
## Income
## 6.218169
```

The Income slope for students is:

```
coef(fit_interaction)["Income"] +
coef(fit_interaction)["Income:StudentYes"]
```

```
## Income
## 4.219018
```

The p-value for `Income:StudentYes` tests whether the Income slopes differ between students and non-students.

Practice Exercise

1. Interpretation

Use the following model:

```
fit_A <- lm(
  Balance ~ Income + Rating + Student,
  data = Credit
)

summary(fit_A)
```

```
##
## Call:
## lm(formula = Balance ~ Income + Rating + Student, data = Credit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -226.126  -80.445   -5.018   65.192  293.234
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -581.07889    13.83463  -42.00  <2e-16 ***
## Income       -7.87493     0.24021  -32.78  <2e-16 ***
## Rating        3.98747     0.05471   72.89  <2e-16 ***
## StudentYes   418.76028    17.23025   24.30  <2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 103.3 on 396 degrees of freedom
## Multiple R-squared:  0.9499, Adjusted R-squared:  0.9495
## F-statistic: 2502 on 3 and 396 DF,  p-value: < 2.2e-16
```

1. What is the estimated coefficient for StudentYes?
2. Interpret it in one sentence.
3. Is it statistically significant at the 5% level?

2. Interaction

```
fit_B <- lm(
  Balance ~ Income * Student,
  data = Credit
)
```

```
summary(fit_B)
```

```
##
## Call:
## lm(formula = Balance ~ Income * Student, data = Credit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -773.39 -325.70 -41.13  321.65  814.04
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    200.6232     33.6984   5.953 5.79e-09 ***
## Income           6.2182       0.5921  10.502 < 2e-16 ***
## StudentYes     476.6758    104.3512   4.568 6.59e-06 ***
## Income:StudentYes -1.9992       1.7313  -1.155  0.249
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 391.6 on 396 degrees of freedom
## Multiple R-squared:  0.2799, Adjusted R-squared:  0.2744
## F-statistic: 51.3 on 3 and 396 DF,  p-value: < 2.2e-16
```

1. What is the Income slope for non-students?
2. What is the Income slope for students?
3. Is there evidence that the slopes differ?

3. Prediction

Use the multiple regression model to predict Balance for two individuals:

- Person 1: Income = 40, Rating = 300, Student = No;
- Person 2: Income = 40, Rating = 300, Student = Yes.

```
new_people <- data.frame(
  Income = c(40, 40),
  Rating = c(300, 300),
  Student = c("No", "Yes")
)
```

```
)  
  
predict(  
  fit_A,  
  newdata = new_people,  
  interval = "confidence",  
  level = 0.95  
)
```

```
##           fit           lwr           upr  
## 1 300.1656 288.6276 311.7036  
## 2 718.9259 686.5934 751.2584
```

Why do the two individuals have different predicted Balance values even though they have the same Income and Rating?